



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 54 OF 84

## ACCEPTANCE TEST REPORT

**Airbus A350F Motor Control Unit 2 (MCU2)**

**CWC P/N 190701-1**

Serial number of MCU2: \_\_\_\_\_

Revision letter of Assembly Part List: \_\_\_\_\_

Test equipment calibration check: \_\_\_\_\_ Pass Fail

Test software no.: \_\_\_\_\_ Rev.

Test performed by: \_\_\_\_\_ (signature)

Test data verified by: \_\_\_\_\_ (Signature)

Start date: \_\_\_\_\_ End Date: \_\_\_\_\_

Environmental conditions: \_\_\_\_\_ Temperature \_\_\_\_°C Humidity \_\_\_\_% Relative

### REMARKS:



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 55 OF 84

Motor Control Unit 2 P/N 190701-1

S/N \_\_\_\_\_

**Ground Bonding  
Test 5.1**

Date: \_\_\_\_\_

Ground Bonding Resistance

Pass \_\_\_\_\_ Fail \_\_\_\_\_

**Test Data**

| Location   | Measured Resistance Under Load (mΩ) | Limit  | Pass/Fail |
|------------|-------------------------------------|--------|-----------|
| 1-2        |                                     | ≤10 mΩ |           |
| 1-3        |                                     | ≤10 mΩ |           |
| 1-4        |                                     | ≤10 mΩ |           |
| 1-5        |                                     | ≤10 mΩ |           |
| 6-2        |                                     | ≤10 mΩ |           |
| 6-3        |                                     | ≤10 mΩ |           |
| 6-4        |                                     | ≤10 mΩ |           |
| 6-5        |                                     | ≤10 mΩ |           |
| J1 Pin D-2 |                                     | ≤20 mΩ |           |
| J1 Pin D-3 |                                     | ≤20 mΩ |           |
| J1 Pin D-4 |                                     | ≤20 mΩ |           |
| J1 Pin D-5 |                                     | ≤20 mΩ |           |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 56 OF 84

Motor Control Unit 2 P/N 190701-1

S/N \_\_\_\_\_

**Dielectric Test**  
**Test 5.2**

Date: \_\_\_\_\_

**OVERALL**

Pass \_\_\_\_\_ Fail \_\_\_\_\_

Leakage Current ( $\leq 1$  mA)

Pass \_\_\_\_\_ Fail \_\_\_\_\_

Visual Inspection (No damage to Location 1)

Pass \_\_\_\_\_ Fail \_\_\_\_\_

**Test Data –**

**Leakage Current (mA)**

|        | TG1      | TG2     | TG3 |     |     |     |
|--------|----------|---------|-----|-----|-----|-----|
| CH_GND | 1500 VAC | 500 VDC |     | TG1 | TG2 | TG3 |
| TG1    |          |         |     |     |     |     |
| TG2    | 500 VDC  |         |     |     |     |     |
| TG3    | 500 VDC  | 500 VDC |     |     |     |     |
| CH_GND |          |         |     |     |     |     |
| TG1    |          |         |     |     |     |     |
| TG2    |          |         |     |     |     |     |
| TG3    |          |         |     |     |     |     |

Instances of insulation breakdown or failure:



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 57 OF 84

Motor Control Unit 2 P/N 190701-1 S/N \_\_\_\_\_

**Galvanic Insulation Resistance**  
**Test 5.3**

Date: \_\_\_\_\_

**OVERALL**

Pass \_\_\_\_\_ Fail \_\_\_\_\_

Baseline Insulation Resistance (≥10 MΩ)

Pass \_\_\_\_\_ Fail \_\_\_\_\_

Visual Inspection (no damage to Location 1)

Pass \_\_\_\_\_ Fail \_\_\_\_\_

**Test Data – Insulation Resistance (MΩ)**

|        | TG1     | TG2    | TG3 |     |     |     |
|--------|---------|--------|-----|-----|-----|-----|
| CH_GND | 500 VDC | 45 VDC |     |     |     |     |
| TG1    |         |        |     | TG1 | TG2 | TG3 |
| TG2    | 45 VDC  |        |     |     |     |     |
| TG3    | 45 VDC  | 45 VDC |     |     |     |     |
| CH_GND |         |        |     |     |     |     |
| TG1    |         |        |     |     |     |     |
| TG2    |         |        |     |     |     |     |
| TG3    |         |        |     |     |     |     |

Any observed insulation resistance anomalies or breakdowns:



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 58 OF 84

Motor Control Unit 2 P/N 190701-1 S/N \_\_\_\_\_

**Test of Presence (Breakdown Voltage) of Lightning Protection Devices**  
**Test 5.4**

Date: \_\_\_\_\_

**OVERALL**

Pass \_\_\_\_\_ Fail \_\_\_\_\_

| Test No. | Pin -     | Pin +     | V_BR          |                     |             |
|----------|-----------|-----------|---------------|---------------------|-------------|
|          |           |           | V_BR Measured | Test Limit, Max (V) | I_Test (mA) |
| 1        | J1 Pin D  | J1 Pin A  |               | 494                 | 1           |
| 2        | J1 Pin D  | J1 Pin B  |               | 494                 | 1           |
| 3        | J1 Pin D  | J1 Pin C  |               | 494                 | 1           |
| 4        | J2 Pin 32 | J2 Pin 17 |               | 7                   | 10          |
| 5        | J2 Pin 32 | J2 Pin 25 |               | 7                   | 10          |
| 6        | J2 Pin 32 | J2 Pin 7  |               | 7                   | 10          |
| 7        | J2 Pin 23 | J2 Pin 39 |               | 7                   | 10          |
| 8        | J2 Pin 23 | J2 Pin 24 |               | 7                   | 10          |
| 9        | J2 Pin 23 | J2 Pin 16 |               | 7                   | 10          |
| 10       | J2 Pin 8  | J2 Pin 3  |               | 7                   | 10          |
| 11       | J2 Pin 15 | J2 Pin 9  |               | 7                   | 10          |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 59 OF 84

Motor Control Unit 2 P/N 190701-1 S/N \_\_\_\_\_

**Pin-to-pin Resistance Test**  
**Test 5.5**

Date: \_\_\_\_\_

Resistance Limits ( $\geq 10\text{ M}\Omega$ ) Pass \_\_\_\_ Fail \_\_\_\_

**Test Data**

| Isolation Group 1 |          |                             |                                  |
|-------------------|----------|-----------------------------|----------------------------------|
| TP+               | TP-      | Min Resistance ( $\Omega$ ) | Measured Resistance ( $\Omega$ ) |
| J1 Pin A          | J1 Pin B | 10M                         |                                  |
|                   | J1 Pin C | 10M                         |                                  |
|                   | J1 Pin D | 10M                         |                                  |
|                   | J3 Pin A | 10M                         |                                  |
|                   | J3 Pin B | 10M                         |                                  |
|                   | J3 Pin C | 10M                         |                                  |
|                   | J4 Pin 1 | 10M                         |                                  |
|                   | J4 Pin 2 | 10M                         |                                  |
|                   | J4 Pin 3 | 10M                         |                                  |
| J1 Pin B          | J1 Pin C | 10M                         |                                  |
|                   | J1 Pin D | 10M                         |                                  |
|                   | J3 Pin A | 10M                         |                                  |
|                   | J3 Pin B | 10M                         |                                  |
|                   | J3 Pin C | 10M                         |                                  |
|                   | J4 Pin 1 | 10M                         |                                  |
|                   | J4 Pin 2 | 10M                         |                                  |
|                   | J4 Pin 3 | 10M                         |                                  |
| J1 Pin C          | J1 Pin D | 10M                         |                                  |
|                   | J3 Pin A | 10M                         |                                  |
|                   | J3 Pin B | 10M                         |                                  |
|                   | J3 Pin C | 10M                         |                                  |
|                   | J4 Pin 1 | 10M                         |                                  |
|                   | J4 Pin 2 | 10M                         |                                  |
|                   | J4 Pin 3 | 10M                         |                                  |
| J1 Pin D          | J3 Pin A | 10M                         |                                  |
|                   | J3 Pin B | 10M                         |                                  |
|                   | J3 Pin C | 10M                         |                                  |
|                   | J4 Pin 1 | 10M                         |                                  |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 60 OF 84

|                   |           |                             |                                  |
|-------------------|-----------|-----------------------------|----------------------------------|
| J3 Pin A          | J4 Pin 2  | 10M                         |                                  |
|                   | J4 Pin 3  | 10M                         |                                  |
|                   | J3 Pin B  | 10M                         |                                  |
|                   | J3 Pin C  | 10M                         |                                  |
|                   | J4 Pin 1  | 10M                         |                                  |
|                   | J4 Pin 2  | 10M                         |                                  |
| J3 Pin B          | J4 Pin 3  | 10M                         |                                  |
|                   | J3 Pin C  | 10M                         |                                  |
|                   | J4 Pin 1  | 10M                         |                                  |
|                   | J4 Pin 2  | 10M                         |                                  |
| J3 Pin C          | J4 Pin 3  | 10M                         |                                  |
|                   | J4 Pin 1  | 10M                         |                                  |
|                   | J4 Pin 2  | 10M                         |                                  |
| J4 Pin 1          | J4 Pin 3  | 10M                         |                                  |
|                   | J4 Pin 2  | 10M                         |                                  |
| J4 Pin 2          | J4 Pin 3  | 10M                         |                                  |
| Isolation Group 2 |           |                             |                                  |
| TP+               | TP-       | Min Resistance ( $\Omega$ ) | Measured Resistance ( $\Omega$ ) |
| J2 Pin 17         | J2 Pin 25 | 10M                         |                                  |
|                   | J2 Pin 32 | 10M                         |                                  |
|                   | J2 Pin 4  | 10M                         |                                  |
|                   | J2 Pin 11 | 10M                         |                                  |
|                   | J2 Pin 10 | 10M                         |                                  |
|                   | J2 Pin 18 | 10M                         |                                  |
| J2 Pin 25         | J2 Pin 32 | 10M                         |                                  |
|                   | J2 Pin 4  | 10M                         |                                  |
|                   | J2 Pin 11 | 10M                         |                                  |
|                   | J2 Pin 10 | 10M                         |                                  |
|                   | J2 Pin 18 | 10M                         |                                  |
| J2 Pin 32         | J2 Pin 4  | 10M                         |                                  |
|                   | J2 Pin 11 | 10M                         |                                  |
|                   | J2 Pin 10 | 10M                         |                                  |
|                   | J2 Pin 18 | 10M                         |                                  |
| J2 Pin 4          | J2 Pin 11 | 10M                         |                                  |
|                   | J2 Pin 10 | 10M                         |                                  |
|                   | J2 Pin 18 | 10M                         |                                  |
| J2 Pin 11         | J2 Pin 10 | 10M                         |                                  |
|                   | J2 Pin 18 | 10M                         |                                  |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 61 OF 84

|                          |            |                                             |                                                  |
|--------------------------|------------|---------------------------------------------|--------------------------------------------------|
| J2 Pin 10                | J2 Pin 18  | 10M                                         |                                                  |
|                          |            |                                             |                                                  |
| <b>Isolation Group 3</b> |            |                                             |                                                  |
| <b>TP+</b>               | <b>TP-</b> | <b>Min Resistance (<math>\Omega</math>)</b> | <b>Measured Resistance (<math>\Omega</math>)</b> |
| J2 Pin 47                | J2 Pin 41  | 10M                                         |                                                  |
|                          | J2 Pin 49  | 10M                                         |                                                  |
|                          | J2 Pin 50  | 10M                                         |                                                  |
|                          | J2 Pin 53  | 10M                                         |                                                  |
|                          | J2 Pin 54  | 10M                                         |                                                  |
|                          | J2 Pin 55  | 10M                                         |                                                  |
|                          | J2 Pin 42  | 10M                                         |                                                  |
|                          | J2 Pin 34  | 10M                                         |                                                  |
|                          | J2 Pin 48  | 10M                                         |                                                  |
|                          | J2 Pin 38  | 10M                                         |                                                  |
|                          | J2 Pin 52  | 10M                                         |                                                  |
|                          | J2 Pin 46  | 10M                                         |                                                  |
|                          | J2 Pin 1   | 10M                                         |                                                  |
|                          | J2 Pin 2   | 10M                                         |                                                  |
|                          | J2 Pin 31  | 10M                                         |                                                  |
|                          | J2 Pin 23  | 10M                                         |                                                  |
|                          | J2 Pin 39  | 10M                                         |                                                  |
|                          | J2 Pin 24  | 10M                                         |                                                  |
|                          | J2 Pin 16  | 10M                                         |                                                  |
|                          | J2 Pin 7   | 10M                                         |                                                  |
|                          | J2 Pin 8   | 10M                                         |                                                  |
|                          | J2 Pin 3   | 10M                                         |                                                  |
|                          | J2 Pin 14  | 10M                                         |                                                  |
|                          | J2 Pin 15  | 10M                                         |                                                  |
|                          | J2 Pin 9   | 10M                                         |                                                  |
| J2 Pin 41                | J2 Pin 49  | 10M                                         |                                                  |
|                          | J2 Pin 50  | 10M                                         |                                                  |
|                          | J2 Pin 53  | 10M                                         |                                                  |
|                          | J2 Pin 54  | 10M                                         |                                                  |
|                          | J2 Pin 55  | 10M                                         |                                                  |
|                          | J2 Pin 42  | 10M                                         |                                                  |
|                          | J2 Pin 34  | 10M                                         |                                                  |
|                          | J2 Pin 48  | 10M                                         |                                                  |
|                          | J2 Pin 38  | 10M                                         |                                                  |
|                          | J2 Pin 52  | 10M                                         |                                                  |
|                          | J2 Pin 46  | 10M                                         |                                                  |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 62 OF 84

|           |           |     |  |
|-----------|-----------|-----|--|
|           | J2 Pin 1  | 10M |  |
|           | J2 Pin 2  | 10M |  |
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 49 | J2 Pin 50 | 10M |  |
|           | J2 Pin 53 | 10M |  |
|           | J2 Pin 54 | 10M |  |
|           | J2 Pin 55 | 10M |  |
|           | J2 Pin 42 | 10M |  |
|           | J2 Pin 34 | 10M |  |
|           | J2 Pin 48 | 10M |  |
|           | J2 Pin 38 | 10M |  |
|           | J2 Pin 52 | 10M |  |
|           | J2 Pin 46 | 10M |  |
|           | J2 Pin 1  | 10M |  |
|           | J2 Pin 2  | 10M |  |
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 50 | J2 Pin 53 | 10M |  |
|           | J2 Pin 54 | 10M |  |
|           | J2 Pin 55 | 10M |  |
|           | J2 Pin 42 | 10M |  |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 63 OF 84

|           |           |     |  |
|-----------|-----------|-----|--|
|           | J2 Pin 34 | 10M |  |
|           | J2 Pin 48 | 10M |  |
|           | J2 Pin 38 | 10M |  |
|           | J2 Pin 52 | 10M |  |
|           | J2 Pin 46 | 10M |  |
|           | J2 Pin 1  | 10M |  |
|           | J2 Pin 2  | 10M |  |
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 53 | J2 Pin 54 | 10M |  |
|           | J2 Pin 55 | 10M |  |
|           | J2 Pin 42 | 10M |  |
|           | J2 Pin 34 | 10M |  |
|           | J2 Pin 48 | 10M |  |
|           | J2 Pin 38 | 10M |  |
|           | J2 Pin 52 | 10M |  |
|           | J2 Pin 46 | 10M |  |
|           | J2 Pin 1  | 10M |  |
|           | J2 Pin 2  | 10M |  |
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 54 | J2 Pin 55 | 10M |  |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 64 OF 84

|           |           |     |  |
|-----------|-----------|-----|--|
|           | J2 Pin 42 | 10M |  |
|           | J2 Pin 34 | 10M |  |
|           | J2 Pin 48 | 10M |  |
|           | J2 Pin 38 | 10M |  |
|           | J2 Pin 52 | 10M |  |
|           | J2 Pin 46 | 10M |  |
|           | J2 Pin 1  | 10M |  |
|           | J2 Pin 2  | 10M |  |
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 55 | J2 Pin 42 | 10M |  |
|           | J2 Pin 34 | 10M |  |
|           | J2 Pin 48 | 10M |  |
|           | J2 Pin 38 | 10M |  |
|           | J2 Pin 52 | 10M |  |
|           | J2 Pin 46 | 10M |  |
|           | J2 Pin 1  | 10M |  |
|           | J2 Pin 2  | 10M |  |
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
| J2 Pin 42 | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
|           | J2 Pin 42 | 10M |  |
|           | J2 Pin 34 | 10M |  |
|           | J2 Pin 48 | 10M |  |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 65 OF 84

|           |           |     |  |
|-----------|-----------|-----|--|
|           | J2 Pin 38 | 10M |  |
|           | J2 Pin 52 | 10M |  |
|           | J2 Pin 46 | 10M |  |
|           | J2 Pin 1  | 10M |  |
|           | J2 Pin 2  | 10M |  |
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 34 | J2 Pin 48 | 10M |  |
|           | J2 Pin 38 | 10M |  |
|           | J2 Pin 52 | 10M |  |
|           | J2 Pin 46 | 10M |  |
|           | J2 Pin 1  | 10M |  |
|           | J2 Pin 2  | 10M |  |
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 48 | J2 Pin 38 | 10M |  |
|           | J2 Pin 52 | 10M |  |
|           | J2 Pin 46 | 10M |  |
|           | J2 Pin 1  | 10M |  |
|           | J2 Pin 2  | 10M |  |
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 66 OF 84

|           |           |     |  |
|-----------|-----------|-----|--|
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 38 | J2 Pin 52 | 10M |  |
|           | J2 Pin 46 | 10M |  |
|           | J2 Pin 1  | 10M |  |
|           | J2 Pin 2  | 10M |  |
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 52 | J2 Pin 46 | 10M |  |
|           | J2 Pin 1  | 10M |  |
|           | J2 Pin 2  | 10M |  |
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
| J2 Pin 46 | J2 Pin 1  | 10M |  |
|           | J2 Pin 2  | 10M |  |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 67 OF 84

|           |           |     |  |
|-----------|-----------|-----|--|
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 1  | J2 Pin 2  | 10M |  |
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
| J2 Pin 2  | J2 Pin 9  | 10M |  |
|           | J2 Pin 31 | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
| J2 Pin 31 | J2 Pin 9  | 10M |  |
|           | J2 Pin 23 | 10M |  |
|           | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 68 OF 84

|           |           |     |  |
|-----------|-----------|-----|--|
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 23 | J2 Pin 39 | 10M |  |
|           | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
| J2 Pin 39 | J2 Pin 24 | 10M |  |
|           | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 24 | J2 Pin 16 | 10M |  |
|           | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 16 | J2 Pin 7  | 10M |  |
|           | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 7  | J2 Pin 8  | 10M |  |
|           | J2 Pin 3  | 10M |  |
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 9  | J2 Pin 3  | 10M |  |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 69 OF 84

|           |           |     |  |
|-----------|-----------|-----|--|
|           | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 3  | J2 Pin 14 | 10M |  |
|           | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 14 | J2 Pin 15 | 10M |  |
|           | J2 Pin 9  | 10M |  |
| J2 Pin 15 | J2 Pin 9  | 10M |  |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 70 OF 84

Motor Control Unit 2 P/N 190701-1

S/N \_\_\_\_\_

**Inrush Current Limit Test**  
**Test 5.6**

Inrush current envelope

Pass \_\_\_\_\_ Fail \_\_\_\_\_

**Test Data**



Note: this is an example of the data to be displayed / recorded (not the pass/fail envelope).



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 71 OF 84

Motor Control Unit 2 P/N 190701-1 S/N \_\_\_\_\_

**Standby Current Draw Test**  
**Test 5.7**

Date: \_\_\_\_\_

**OVERALL**

**Pass**\_\_\_\_\_ **Fail** \_\_\_\_\_

**Test Data**

| Measurement     | Recorded Value | Pass/Fail Limit | Pass/Fail |
|-----------------|----------------|-----------------|-----------|
| Phase A current |                | <1A             |           |
| Phase B current |                | <1A             |           |
| Phase C current |                | <1A             |           |
| Health Status   |                | READY           |           |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 72 OF 84

Motor Control Unit 2 P/N 190701-1 S/N \_\_\_\_\_

**Input/Output (I/O) Range Test – ENABLE and STATUS Discretes**  
**Test 5.8**

Date: \_\_\_\_\_

I/O signals specified ranges

Pass \_\_\_\_ Fail \_\_\_\_

**Test Data**

| Measurement                                      | Recorded Value | Pass/Fail Limit                                  | Pass/Fail |
|--------------------------------------------------|----------------|--------------------------------------------------|-----------|
| $V_{\text{threshold\_up}}$                       |                | $\geq 0.9 \text{ V}$                             |           |
| $V_{\text{threshold\_down}}$                     |                | $\leq 1.2 \text{ V}$                             |           |
| Voltage on J2 Pin 25 w/ ENABLE signal is present |                | $4.4 \text{ V} \leq \text{STS} \leq 5 \text{ V}$ |           |
| Voltage on J2 Pin 25 NO ENABLE signal is present |                | $< 0.2 \text{ V}$                                |           |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 73 OF 84

Motor Control Unit 2 P/N 190701-1 S/N \_\_\_\_\_

Input/Output (I/O) Range Test – ARINC 429  
Test 5.9

Date: \_\_\_\_\_

Data within specified ranges

Pass \_\_\_\_ Fail \_\_\_\_

Test Data

| Measurement               | Recorded Value | Pass/Fail Limit          | Pass/Fail |
|---------------------------|----------------|--------------------------|-----------|
| A429_MCU2_TX High Voltage |                | Voltage $\geq 9$ V       |           |
| A429_MCU2_TX Low Voltage  |                | Voltage $\leq -9$ V      |           |
| Slew Rate                 |                | 4-8 V $\mu\text{s}^{-1}$ |           |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 74 OF 84

Motor Control Unit 2 P/N 190701-1 S/N \_\_\_\_\_

Input/Output Range Test – NEOT Hall Effect Sensors  
Test 5.10

Date: \_\_\_\_\_

OVERALL

Pass \_\_\_\_ Fail \_\_\_\_

Test Data

| Test No. | Hall Effect Switch Position |              | ARINC STATUS | Required ARINC STATUS     | Pass/ Fail |
|----------|-----------------------------|--------------|--------------|---------------------------|------------|
|          | NEOT_LAA_RET                | NEOT_LAA_EXT |              |                           |            |
| 1        | OPEN                        | CLOSED       |              | LAA_STHS_ACTIVE_EXT ENDED |            |
| 2        | CLOSED                      | OPEN         |              | LAA_STHS_ACTIVE_RETRACTED |            |
| -        | NEOT_LOA_RET                | NEOT_LOA_EXT | LOA          |                           |            |
| 3        | OPEN                        | CLOSED       |              | LOA_STHS_ACTIVE_EXTENDED  |            |
| 4        | CLOSED                      | OPEN         |              | LOA_STHS_ACTIVE_RETRACTED |            |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 75 OF 84

Motor Control Unit 2 P/N 190701-1 S/N \_\_\_\_\_

Input/Output Range Test – LAA and LOA Temperature  
Test 5.11

Date: \_\_\_\_\_

OVERALL

Pass \_\_\_\_ Fail \_\_\_\_

Test Data

| ARINC FAULT   | Temperature | Fault Status | Pass / Fail Criteria | Pass/Fail |
|---------------|-------------|--------------|----------------------|-----------|
| LAA_OVER_TEMP | 105°C       |              | READY                |           |
|               | 115°C       |              | FAULT                |           |
|               | 90°C        |              | FAULT                |           |
|               | 80°C        |              | READY                |           |
| LOA_OVER_TEMP | 105°C       |              | READY                |           |
|               | 115°C       |              | FAULT                |           |
|               | 90°C        |              | FAULT                |           |
|               | 80°C        |              | READY                |           |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 76 OF 84

Motor Control Unit 2 P/N 190701-1

S/N \_\_\_\_\_

**Motor Output Test**  
**Test 5.12**

OVERALL

Pass \_\_\_\_\_ Fail \_\_\_\_\_

**Test Data**

| MCU2 Enabled | MCU2 Status | Command Issued               | Output Current (Arms) | Pass/Fail Criteria | Pass/Fail |
|--------------|-------------|------------------------------|-----------------------|--------------------|-----------|
| No           | NOT READY   |                              |                       | < 50mArms          |           |
| No           | NOT READY   | LAA <input type="checkbox"/> |                       | < 50mArms          |           |
| Yes          | READY       |                              |                       | < 50mArms          |           |
| Yes          | READY       | LAA <input type="checkbox"/> |                       | > 1Arms            |           |
| Yes          | READY       |                              |                       | < 50mArms          |           |
| No           | NOT READY   |                              |                       | < 50mArms          |           |
| No           | NOT READY   | LAA <input type="checkbox"/> |                       | < 50mArms          |           |
| No           | NOT READY   |                              |                       | < 50mArms          |           |
| No           | NOT READY   | LOA <input type="checkbox"/> |                       | < 50mArms          |           |
| Yes          | READY       |                              |                       | < 50mArms          |           |
| Yes          | READY       | LOA <input type="checkbox"/> |                       | > 100mArms         |           |
| Yes          | READY       |                              |                       | < 50mArms          |           |
| No           | NOT READY   |                              |                       | < 50mArms          |           |
| No           | NOT READY   | LOA <input type="checkbox"/> |                       | < 50mArms          |           |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 77 OF 84

Motor Control Unit 2 P/N 190701-1

S/N \_\_\_\_\_

**Interlock Test**  
**Test 5.13**

**OVERALL**

**Pass** \_\_\_\_\_ **Fail** \_\_\_\_\_

**Test Data**

| Test No. | ULAA | LAA | ULOA | LOA | Valid | Fault Indicated | Motor Operates | Pass/Fail |
|----------|------|-----|------|-----|-------|-----------------|----------------|-----------|
| 1        | 0    | 0   | 1    | 1   | No    | Yes             |                |           |
| 2        | 0    | 1   | 0    | 1   | No    | Yes             |                |           |
| 3        | 0    | 1   | 1    | 0   | No    | Yes             |                |           |
| 4        | 0    | 1   | 1    | 1   | No    | Yes             |                |           |
| 5        | 1    | 0   | 0    | 1   | No    | Yes             |                |           |
| 6        | 1    | 0   | 1    | 0   | No    | Yes             |                |           |
| 7        | 1    | 0   | 1    | 1   | No    | Yes             |                |           |
| 8        | 1    | 1   | 0    | 0   | No    | Yes             |                |           |
| 9        | 1    | 1   | 0    | 1   | No    | Yes             |                |           |
| 10       | 1    | 1   | 1    | 0   | No    | Yes             |                |           |
| 11       | 1    | 1   | 1    | 1   | No    | Yes             |                |           |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 78 OF 84

Motor Control Unit 2 P/N 190701-1

S/N \_\_\_\_\_

**Efficiency Test**  
**Test 5.14**

Time: \_\_\_\_\_

Date: \_\_\_\_\_

**Efficiency** (≥85%)

**Pass** \_\_\_\_ **Fail** \_\_\_\_

**Test Data**

| Test  | Output Load (kW) | Input Load (kW) | Efficiency (%) | Pass/Fail Criteria | Pass/Fail |
|-------|------------------|-----------------|----------------|--------------------|-----------|
| Latch |                  |                 |                | ≥60%               |           |
| Lock  |                  |                 |                | ≥40%               |           |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 79 OF 84

**Functional Test**  
**Test 5.15**

Time: \_\_\_\_\_

Date: \_\_\_\_\_

**OVERALL**

Pass \_\_\_\_\_ Fail \_\_\_\_\_

**Test Data - Unloaded Speed**

| Test | Command | Speed | Unloaded Steady State Speed | Pass/Fail Limit (rpm) | Pass/Fail |
|------|---------|-------|-----------------------------|-----------------------|-----------|
| 1    | LATCH   | High  |                             | 9785 / 9215           |           |
| 2    | LATCH   | Low   |                             | 649 / 611             |           |
| 3    | UNLATCH | High  |                             | 9785 / 9215           |           |
| 4    | UNLATCH | Low   |                             | 9785 / 9215           |           |
| 5    | LOCK    | High  |                             | 9425 / 8875           |           |
| 6    | LOCK    | Low   |                             | 629 / 591             |           |
| 7    | UNLOCK  | High  |                             | 9425 / 8875           |           |
| 8    | UNLOCK  | Low   |                             | 629 / 591             |           |

**Test Data - Loaded Torque**

| Test | Command | Speed | Dyno Speed Set Point (rpm) | Torque at Set Speed (Nm) | Pass/Fail Limit (Nm) | Pass/Fail |
|------|---------|-------|----------------------------|--------------------------|----------------------|-----------|
| 1    | LATCH   | High  | 8550                       |                          | 0.614 / 0.508        |           |
| 2    | LATCH   | Low   | 567                        |                          | 0.178 / 0.120        |           |
| 3    | UNLATCH | High  | 8550                       |                          | 0.614 / 0.508        |           |
| 4    | UNLATCH | Low   | 567                        |                          | 0.178 / 0.120        |           |
| 5    | LOCK    | High  | 8235                       |                          | 0.063 / 0.052        |           |
| 6    | LOCK    | Low   | 549                        |                          | 0.032 / 0.026        |           |
| 7    | UNLOCK  | High  | 8235                       |                          | 0.063 / 0.052        |           |
| 8    | UNLOCK  | Low   | 549                        |                          | 0.032 / 0.026        |           |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 80 OF 84

### Test Data - Loaded Current

| Test | Command | Speed | Dyno Speed Set Point (rpm) | Current at Set Speed (A) | Pass/Fail Limit (A) | Pass/Fail |
|------|---------|-------|----------------------------|--------------------------|---------------------|-----------|
| 1    | LATCH   | High  | 8550                       |                          | 3.28 / 2.56         |           |
| 2    | LATCH   | Low   | 567                        |                          | 0.77 / 0.54         |           |
| 3    | UNLATCH | High  | 8550                       |                          | 3.28 / 2.56         |           |
| 4    | UNLATCH | Low   | 567                        |                          | 0.77 / 0.54         |           |
| 5    | LOCK    | High  | 8235                       |                          | 0.38 / 0.29         |           |
| 6    | LOCK    | Low   | 549                        |                          | 0.19 / 0.15         |           |
| 7    | UNLOCK  | High  | 8235                       |                          | 0.38 / 0.29         |           |
| 8    | UNLOCK  | Low   | 549                        |                          | 0.19 / 0.15         |           |

### Test Data - Jam Current and Torque

| Test | Command | Speed | Jam Torque (Nm) | Jam Torque Pass/Fail Limit (Nm) | Jam Current (A) | Jam Current Pass/Fail Limit | Pass/Fail |
|------|---------|-------|-----------------|---------------------------------|-----------------|-----------------------------|-----------|
| 1    | LATCH   | High  |                 | 0.669 / 0.476                   |                 | 4.00 / 3.63                 |           |
| 2    | LATCH   | Low   |                 | 0.178 / 0.127                   |                 | 0.935 / 0.77                |           |
| 3    | UNLATCH | High  |                 | 0.669 / 0.476                   |                 | 4.00 / 3.63                 |           |
| 4    | UNLATCH | Low   |                 | 0.178 / 0.127                   |                 | 0.935 / 0.77                |           |
| 5    | LOCK    | High  |                 | 0.077 / 0.055                   |                 | 0.46 / 0.417                |           |
| 6    | LOCK    | Low   |                 | 0.039 / 0.028                   |                 | 0.23 / 0.208                |           |
| 7    | UNLOCK  | High  |                 | 0.077 / 0.055                   |                 | 0.46 / 0.417                |           |
| 8    | UNLOCK  | Low   |                 | 0.039 / 0.028                   |                 | 0.23 / 0.208                |           |

### Test Data - Jam Fault Status

| Test | Command | Speed | Jam Fault Status | Jam Fault Status Pass/Fail | Pass/Fail |
|------|---------|-------|------------------|----------------------------|-----------|
| 1    | LATCH   | High  |                  | Fault Active               |           |
| 2    | LATCH   | Low   |                  | Fault Active               |           |
| 3    | UNLATCH | High  |                  | Fault Active               |           |
| 4    | UNLATCH | Low   |                  | Fault Active               |           |
| 5    | LOCK    | High  |                  | Fault Active               |           |
| 6    | LOCK    | Low   |                  | Fault Active               |           |
| 7    | UNLOCK  | High  |                  | Fault Active               |           |
| 8    | UNLOCK  | Low   |                  | Fault Active               |           |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 81 OF 84

Motor Control Unit 2 P/N 190701-1 S/N \_\_\_\_\_

**Jam Fault Detection Test**  
**Test 5.16**

Time: \_\_\_\_\_

Date: \_\_\_\_\_

**OVERALL**

**Pass** \_\_\_\_\_ **Fail** \_\_\_\_\_

**Test Data**

| Command | Time (s) | Criteria | Output Phase Current Before Fault (Arms) | Output Phase Current After Fault (Arms) | Pass/Fail |
|---------|----------|----------|------------------------------------------|-----------------------------------------|-----------|
| LAA     |          | 3–5.5 s  |                                          |                                         |           |
| LOA     |          |          |                                          |                                         |           |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 82 OF 84

Motor Control Unit 2 P/N 190701-1 S/N \_\_\_\_\_

**Non-zero Motion Flag Test**  
**Test 5.17**

Time: \_\_\_\_\_

Date: \_\_\_\_\_

**OVERALL**

Pass \_\_\_\_\_ Fail \_\_\_\_\_

**Test Data**

| Condition           | NON_Zero_Flag<br>Required Status | NON_Zero_Flag<br>Actual Status | Pass/Fail |
|---------------------|----------------------------------|--------------------------------|-----------|
| LAA – no command    | Inactive                         |                                |           |
| LATCH (Low Speed)   | Active                           |                                |           |
| LAA – no command    | Inactive                         |                                |           |
| UNLATCH (Low Speed) | Active                           |                                |           |
| LAA – no command    | Inactive                         |                                |           |
| LOA – no command    | Inactive                         |                                |           |
| LOCK (Low Speed)    | Active                           |                                |           |
| LOA – no command    | Inactive                         |                                |           |
| UNLOCK (Low Speed)  | Active                           |                                |           |
| LOA – no command    | Inactive                         |                                |           |



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 83 OF 84

Motor Control Unit 2 P/N 190701-1

S/N \_\_\_\_\_

Date: \_\_\_\_\_

Inspector: \_\_\_\_\_

190701-ICD Rev \_\_\_\_\_

### Marking and Nameplate

#### Test 5.18

| Document | Revision Level |
|----------|----------------|
|          |                |
|          |                |
|          |                |
|          |                |

### OVERALL

Pass \_\_\_\_\_ Fail \_\_\_\_\_

Correct Information (All markings correct) Pass \_\_\_\_\_ Fail \_\_\_\_\_

Proper Placement (Locations correct) Pass \_\_\_\_\_ Fail \_\_\_\_\_

Nameplate Legibility (Clearly legible) Pass \_\_\_\_\_ Fail \_\_\_\_\_

Nameplate Security (No peeling, lifting, or Misalignment.) Pass \_\_\_\_\_ Fail \_\_\_\_\_

### Interface Dimensions

#### Test 5.19

Compliance with ICD Drawings (Interface dimensions within tolerance) Pass \_\_\_\_\_ Fail \_\_\_\_\_

### Weight Check

#### Test 5.20

Target Weight ( $\leq 4$  kg) Pass \_\_\_\_\_ Fail \_\_\_\_\_

Weight: \_\_\_\_\_ kg



ACTUATION GROUP  
SHELBY, NC 28152  
CAGE CODE: 71791

CPS 8282 REV A

SHEET NO. 84 OF 84

**Finish and Workmanship**  
**Test 5.21**

Visual Inspection

(No visible flaws)

Pass \_\_\_\_\_ Fail \_\_\_\_\_

Pin check

(No broken or bent pins)

Pass \_\_\_\_\_ Fail \_\_\_\_\_

Notes: